

# **STA 130A: Mathematical Statistics: Brief Course**

## **Lecture 8: Conditional expectation and Independence**

Dogyoon Song

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# Announcements

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## Homework 3: Due Thu, Oct 15, 11:59 PM PT

- Please submit on time and follow the submission instructions
- Please review the homework problems early so you have time to ask questions if needed
- Feel free to ask questions during lecture, in office hours or post them on Piazza

## Midterm 1 is in class on Mon, Oct 19

- You may bring *one **hand-written** letter-sized ( $8.5 \times 11$  inches), double-sided sheet of paper* with formulas and brief notes
- **Calculator:** Simple (non-graphing) calculators only
- **No textbooks** or other study aids beyond the permitted note sheet and calculator
- **SDC accommodations:** Confirm scheduling with AES online

# Agenda

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## Last time:

- Joint, marginal, and conditional PMFs
- Functions of random variables
- Linearity of expectation

## Today:

- Conditional expectation
- The law of total expectation (a.k.a. total expectation theorem)
- Independence of random variables
  - Factorization of expectation
  - Variance of sums of independent random variables

## Recap: Joint, marginal, and conditional PMFs

### The same joint table as Lecture 7

| $p_{X,Y}(x,y)$ | $y = 0$ | $y = 1$ | $y = 2$ | $p_X(x)$ |
|----------------|---------|---------|---------|----------|
| $x = 0$        | 1/8     | 1/8     | 1/4     | 1/2      |
| $x = 1$        | 1/4     | 1/8     | 1/8     | 1/2      |
| $p_Y(y)$       | 3/8     | 1/4     | 3/8     | 1        |

- Inner cells:  $p_{X,Y}$
- Row sums:  $p_X$
- Column sums:  $p_Y$

### Joint vs marginal vs conditional PMFs

- **Joint:** both  $X = 1$  and  $Y = 2$

$$p_{X,Y}(1,2) = P(X=1, Y=2) = \frac{1}{8}.$$

- **Marginal:**  $Y = 2$ , regardless of  $X$

$$p_Y(2) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}.$$

- **Conditional:**  $X = 1$ , given  $Y = 2$

$$p_{X|Y}(1|2) = \frac{p_{X,Y}(1,2)}{p_Y(2)} = \frac{1/8}{3/8} = \frac{1}{3}.$$

**Marginalizing:** For each  $X = x$ , sum over (irrelevant)  $Y \in \{0, 1, 2\}$

**Conditioning:** Keep the  $Y = 2$  column (only) and divide by its total 3/8

# Illustration: Building a joint PMF sequentially

## Example (Short Q&A session)

Let  $X$  be the number of questions asked in a short Q&A session, and assume

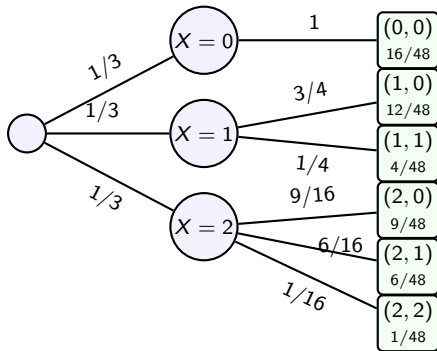
$$P(X = 0) = P(X = 1) = P(X = 2) = \frac{1}{3},$$

Suppose that given  $X = x$ , each question is answered incorrectly with probability  $1/4$ , independently of the others. Let  $Y$  be the number answered incorrectly. Thus

$$Y \mid X = x \sim \text{Binomial}(x, 1/4),$$

and therefore,

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y \mid x).$$



| $p_{X,Y}(x, y)$ | $y = 0$ | $y = 1$ | $y = 2$ |
|-----------------|---------|---------|---------|
| $x = 0$         | 16/48   | 0       | 0       |
| $x = 1$         | 12/48   | 4/48    | 0       |
| $x = 2$         | 9/48    | 6/48    | 1/48    |

# Conditional expectation

## Definition

The **conditional expectation** of  $X$  given an event  $A$  with  $P(A) > 0$  is

$$\mathbb{E}[X \mid A] = \sum_x x p_{X|A}(x), \quad p_{X|A}(x) = P(X = x \mid A).$$

The conditional expectation of  $X$  given a value  $y$  of  $Y$ , with  $p_Y(y) > 0$ , is

$$\mathbb{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y).$$

## Example

| $p_{X,Y}(x, y)$ | $y = 0$ | $y = 1$ | $y = 2$ |
|-----------------|---------|---------|---------|
| $x = 0$         | 1/8     | 1/8     | 1/4     |
| $x = 1$         | 1/4     | 1/8     | 1/8     |

From the joint PMF table

$$\mathbb{E}[X \mid Y = 2] = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{1}{3}.$$

# Law of total expectation

## Theorem

If  $A_1, \dots, A_n$  form a partition of the sample space and  $P(A_i) > 0$  for all  $i$ , then

$$\mathbb{E}[X] = \sum_{i=1}^n P(A_i) \mathbb{E}[X \mid A_i]$$

**Proof.** For each  $x$ ,

$$p_X(x) = \sum_{i=1}^n P(A_i) p_{X|A_i}(x),$$

by the PMF version of the law of total probability. Therefore,

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x p_X(x) = \sum_x x \sum_{i=1}^n P(A_i) p_{X|A_i}(x) = \sum_{i=1}^n P(A_i) \sum_x x p_{X|A_i}(x) \\ &= \sum_{i=1}^n P(A_i) \mathbb{E}[X \mid A_i]. \end{aligned}$$

## Total expectation theorem (cont'd)

### Theorem

For a partition  $A_1, \dots, A_n$  with  $P(A_i) > 0$  for all  $i$ ,

$$\mathbb{E}[X] = \sum_{i=1}^n P(A_i) \mathbb{E}[X \mid A_i].$$

Two useful forms of the same rule:

- For discrete random variables  $X$  and  $Y$ ,

$$\mathbb{E}[X] = \sum_y p_Y(y) \mathbb{E}[X \mid Y = y]$$

- For any event  $B$  with  $P(A_i \cap B) > 0$  for all  $i$ ,

$$\mathbb{E}[X \mid B] = \sum_{i=1}^n P(A_i \mid B) \mathbb{E}[X \mid A_i \cap B]$$

# Total expectation theorem: Toy example

## Example

Let  $X$  be uniform on  $\{0, 1, 2\}$ . Given  $X = x$ , a student guesses independently and uniformly on  $x$  questions, each with four choices and exactly one correct answer. Let  $Y$  count correct answers. Then

$$Y \mid X = x \sim \text{Binomial}(x, 1/4),$$

so

$$\mathbb{E}[Y \mid X = x] = \frac{x}{4}.$$

Therefore,

$$\begin{aligned}\mathbb{E}[Y] &= \sum_x \mathbb{E}[Y \mid X = x] p_X(x) \\ &= \sum_x \frac{x}{4} p_X(x) = \frac{1}{4} \mathbb{E}[X] \\ &= \frac{1}{4} \cdot \frac{0 + 1 + 2}{3} = \frac{1}{4}.\end{aligned}$$

## Exercise for self-study: Portfolio example (1/4)

### Example (State-dependent investment returns)

Suppose the market state after one year is  $M \in \{\text{boom, normal, crash}\}$ , with probabilities

$$P(M = \text{boom}) = 0.20, \quad P(M = \text{normal}) = 0.75, \quad P(M = \text{crash}) = 0.05.$$

Consider two one-year investment options with state-dependent returns:

| Market state           | boom | normal | crash |
|------------------------|------|--------|-------|
| Stock fund ( $R_S$ )   | 30%  | 10%    | -40%  |
| Hedge option ( $R_H$ ) | -10% | -5%    | 100%  |

Assume returns are *fixed within each market state*. Thus the table gives the conditional means:

$$R_S(m) = \mathbb{E}[R_S \mid M = m], \quad R_H(m) = \mathbb{E}[R_H \mid M = m].$$

By the total expectation theorem,

$$\mathbb{E}[R_S] = 0.20 \times 30\% + 0.75 \times 10\% + 0.05 \times (-40\%) = 11.5\%,$$

$$\mathbb{E}[R_H] = 0.20 \times (-10\%) + 0.75 \times (-5\%) + 0.05 \times (100\%) = -0.75\%.$$

## Exercise for self-study: Portfolio example (2/4)

### Example (Maximizing expected return)

Suppose you invest a fraction  $w \in [0, 1]$  in the stock fund and the remaining  $1 - w$  in the hedge.

The portfolio return is

$$R_w = wR_S + (1 - w)R_H.$$

By linearity of expectation,

$$\mathbb{E}[R_w] = w \mathbb{E}[R_S] + (1 - w) \mathbb{E}[R_H] = w \times 11.5\% + (1 - w) \times (-0.75\%).$$

Equivalently,

$$\mathbb{E}[R_w] = 12.25\% w - 0.75\%.$$

Since this is increasing in  $w$ , the expected return is maximized at

$$w = 1.$$

Thus, if the goal is to maximize expected return *alone*, the best choice is to invest everything in the stock:

$$\max_{0 \leq w \leq 1} \mathbb{E}[R_w] = 11.5\%.$$

## Exercise for self-study: Portfolio example (3/4)

### Example (A simple hedging strategy)

Now suppose instead that you want the portfolio return to be *nonnegative in every market state*. Observe

$$R_w = wR_S + (1 - w)R_H = \begin{cases} 40\% w - 10\%, & \text{boom,} \\ 15\% w - 5\%, & \text{normal,} \\ 100\% - 140\% w, & \text{crash.} \end{cases}$$

To keep the return nonnegative in every state, we need

$$40\% w - 10\% \geq 0, \quad 15\% w - 5\% \geq 0, \quad 100\% - 140\% w \geq 0.$$

These inequalities imply

$$w \geq \frac{1}{4}, \quad w \geq \frac{1}{3}, \quad w \leq \frac{5}{7}, \quad \text{and thus,} \quad \frac{1}{3} \leq w \leq \frac{5}{7}$$

Since  $\mathbb{E}[R_w] = 12.25\% w - 0.75\%$  is increasing in  $w$ , the best hedged portfolio is at  $w = \frac{5}{7}$ .

## Exercise for self-study: Portfolio example (4/4)

### Example (A simple hedging strategy)

The portfolio with  $w = \frac{5}{7}$  gives state-by-state returns

$$R_{5/7} = \begin{cases} \frac{13}{70} \approx 18.57\%, & \text{boom,} \\ \frac{2}{35} \approx 5.71\%, & \text{normal,} \\ 0, & \text{crash,} \end{cases}$$

and expected return

$$\mathbb{E}[R_{5/7}] = 12.25\% \times \frac{5}{7} - 0.75\% = 8\%.$$

**Takeaway:** The law of total expectation computes expected returns from market states. A nontrivial portfolio-allocation problem appears once we add a hedging/risk constraint.

## Pop-up quiz

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Which identity correctly expresses the law of total expectation for discrete random variables  $X$  and  $Y$ ?

A)  $\mathbb{E}[X \mid Y = y] = \mathbb{E}[X] p_Y(y)$

B)  $\mathbb{E}[X] = \mathbb{E}[Y \mid X = x]$

C)  $\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y]$

D)  $\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y] p_Y(y)$

**Answer: D.**

The unconditional mean is the weighted average of the conditional means, with weights  $p_Y(y)$ .

**Follow-up:** Why do we need the weights  $p_Y(y)$ ?

# Independence of a random variable from an event

## Definition

Let  $A$  be an event with  $P(A) > 0$ . We say that  $X$  is **independent of**  $A$  if

$$p_{X|A}(x) = p_X(x), \quad \forall x.$$

Equivalently, for every  $x$ , the event  $\{X = x\}$  is independent of  $A$ .

## Example (Two fair die rolls)

Let  $X$  be the first roll of two independent fair dice, and let

$$A = \{\text{the sum of the two rolls is even}\}.$$

For each  $x \in \{1, \dots, 6\}$ , exactly three values of the second roll make the sum even, so

$$P(X = x | A) = \frac{3/36}{18/36} = \frac{1}{6} = p_X(x).$$

# Independence of random variables

## Definition

Discrete random variables  $X$  and  $Y$  are **independent** if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y), \quad \forall (x,y).$$

Equivalently, for every  $y$  with  $p_Y(y) > 0$ ,

$$p_{X|Y}(x | y) = p_X(x), \quad \forall x.$$

- Knowing the value of  $Y$  does not change the distribution of  $X$ .
- The condition must hold for every pair  $(x,y)$ , not just some of them.
- If  $X$  and  $Y$  are independent, then  $g(X)$  and  $h(Y)$  are independent for any functions  $g$  and  $h$ .

## Example (A non-example)

For two independent fair die rolls, let  $X$  be the first result and  $Y$  their sum. Then  $P(X = 6, Y = 12) = P(\{(6,6)\}) = \frac{1}{36}$ , whereas  $P(X = 6)P(Y = 12) = \frac{1}{6} \cdot \frac{1}{36} = \frac{1}{216}$ . Since these are not equal,  $X$  and  $Y$  are not independent.

## Example: Same experiment, yet independent

### Example (Two independent fair die rolls)

Let

$$X = \mathbf{1}_{\{\text{first roll is even}\}}, \quad Y = \mathbf{1}_{\{\text{sum of the two rolls is even}\}}.$$

Then  $X, Y \in \{0, 1\}$ , and the joint PMF<sup>a</sup> is

| $p_{X,Y}(x,y)$ | $y = 0$ | $y = 1$ |
|----------------|---------|---------|
| $x = 0$        | 1/4     | 1/4     |
| $x = 1$        | 1/4     | 1/4     |

Since  $p_X(0) = p_X(1) = 1/2$  and  $p_Y(0) = p_Y(1) = 1/2$ ,

$$p_{X,Y}(x,y) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = p_X(x)p_Y(y) \quad \text{for all } x, y \in \{0, 1\}.$$

So  $X$  and  $Y$  are independent.

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<sup>a</sup>Even though  $Y$  depends on both rolls, the parity of the sum is still equally likely to be even or odd once the parity of the first roll is fixed.

## Pop-up quiz

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Two independent fair die rolls are performed. Define

$$X = \text{first roll}, \quad A = \{\text{the second roll is even}\}.$$

**Question:** Which statement is correct?

- A)  $X$  is not independent of  $A$ , because they come from the same experiment.
- B)  $X$  is independent of  $A$ , and  $p_{X|A}(x) = 1/6$  for all  $x \in \{1, \dots, 6\}$ .
- C)  $X$  is independent of  $A$ , but only for  $x \in \{1, \dots, 5\}$ .
- D)  $X$  is independent of  $A$  only because the dice are fair.

**Answer: B.**

The rolls are independent, so conditioning on the second roll being even does not change the first-roll distribution. It remains uniform on  $\{1, \dots, 6\}$ .

**Follow-up:** Would the same conclusion hold if  $A = \{\text{sum is } 8\}$ ?

# Independence implies expectation factorization

## Theorem

*If  $X$  and  $Y$  are independent, then*

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

*More generally, for any functions  $g$  and  $h$ ,*

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \mathbb{E}[h(Y)].$$

**Proof:**

$$\begin{aligned}\mathbb{E}[g(X)h(Y)] &= \sum_{x,y} g(x)h(y) p_{X,Y}(x,y) = \sum_{x,y} g(x)h(y) p_X(x)p_Y(y) \\ &= \left( \sum_x g(x)p_X(x) \right) \left( \sum_y h(y)p_Y(y) \right) \\ &= \mathbb{E}[g(X)] \mathbb{E}[h(Y)].\end{aligned}$$

## Example: Factorization in action

### Example (A binomial second moment)

Let  $X_1, \dots, X_n$  be independent Bernoulli( $p$ ) random variables, and let

$$S = X_1 + \dots + X_n.$$

Then

$$S^2 = \sum_{i=1}^n X_i^2 + 2 \sum_{1 \leq i < j \leq n} X_i X_j.$$

Taking expectations gives

$$\mathbb{E}[S^2] = \sum_{i=1}^n \mathbb{E}[X_i^2] + 2 \sum_{1 \leq i < j \leq n} \mathbb{E}[X_i X_j].$$

Since  $X_i^2 = X_i$  and the  $X_i$  are independent,

$$\mathbb{E}[X_i^2] = p, \quad \mathbb{E}[X_i X_j] = \mathbb{E}[X_i] \mathbb{E}[X_j] = p^2 \quad (i \neq j).$$

Therefore,

$$\mathbb{E}[S^2] = np + n(n-1)p^2.$$

## Caution: Factorization alone does not imply independence

If  $X, Y$  are independent, then  $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$

However, the identity  $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$  does *not* by itself imply independence

### Example

Let  $X$  be uniform on  $\{-1, 0, 1\}$ , and let  $Y = X^2$ . Then

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = \frac{2}{3}, \quad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

So

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y].$$

But  $P(Y = 0 \mid X = 0) = 1 \neq 1/3 = P(Y = 0)$ , so  $X$  and  $Y$  are not independent

Factorization of  $\mathbb{E}[XY]$  is a consequence, but not a characterization of independence

# Independence implies variance additivity

## Theorem

If  $X$  and  $Y$  are independent, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

## Proof:

$$\begin{aligned}\text{Var}(X + Y) &= \mathbb{E}[(X + Y)^2] - (\mathbb{E}[X] + \mathbb{E}[Y])^2 \\ &= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] - \mathbb{E}[X]^2 - 2\mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]^2 \\ &= \text{Var}(X) + \text{Var}(Y) + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]).\end{aligned}$$

Noting that if  $X$  and  $Y$  are independent, then  $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$  completes the proof.

More generally, for mutually independent  $X_1, \dots, X_n$ ,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

## Example: Variance of a binomial random variable

### Example (Expectation and variance of a binomial random variable)

Let  $X_1, \dots, X_n$  be independent Bernoulli( $p$ ) random variables, and let

$$X = X_1 + \dots + X_n.$$

Since

$$\text{Var}(X_i) = p(1 - p),$$

the variance-addition rule gives

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p).$$

So for a binomial random variable  $X \sim \text{Binomial}(n, p)$ ,

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

## Example: Mean and variance of the sample mean

### Example (Sample mean of i.i.d. observations)

Let  $X_1, \dots, X_n$  be independent and identically distributed, with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu, \quad \text{and} \quad \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

- Averaging preserves the mean.
- Averaging reduces the variance by a factor of  $n$ .
- This is one mathematical reason why averaging stabilizes noisy observations.

# Same marginals, different dependence

## Example (Same marginals, different dependence)

Let  $X, Y \in \{0, 1\}$ , and suppose both are marginally Bernoulli(1/2).

### Model 1: independent

| $p_{X,Y}(x, y)$ | $y = 0$ | $y = 1$ |
|-----------------|---------|---------|
| $x = 0$         | 1/4     | 1/4     |
| $x = 1$         | 1/4     | 1/4     |

### Model 2: perfectly dependent

| $p_{X,Y}(x, y)$ | $y = 0$ | $y = 1$ |
|-----------------|---------|---------|
| $x = 0$         | 1/2     | 0       |
| $x = 1$         | 0       | 1/2     |

In both models,

$$\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{2}, \quad \mathbb{E}[X + Y] = 1.$$

But

$$\text{Var}(X + Y) = \frac{1}{2} = \text{Var}(X) + \text{Var}(Y) \quad \text{in Model 1}, \quad \text{Var}(X + Y) = 1 \quad \text{in Model 2}.$$

- Marginal PMFs alone do not determine the dependence structure.
- Dependence can significantly change quantities such as  $P(X = Y)$ ,  $\mathbb{E}[XY]$ , and  $\text{Var}(X + Y)$ .

## Pop-up quiz

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Suppose  $X$  and  $Y$  are both marginally Bernoulli( $1/2$ ).

**Model 1:**  $X$  and  $Y$  are independent.

**Model 2:**  $Y = X$ .

**Question:** Which quantity is the same in both models?

- A)  $\mathbb{E}[XY]$
- B)  $\text{Var}(X + Y)$
- C)  $P(X = Y)$
- D)  $\mathbb{E}[X + Y]$

**Answer: D.**

In both models,  $\mathbb{E}[X] = \mathbb{E}[Y] = 1/2$ , so  $\mathbb{E}[X + Y] = 1$ ; the other quantities depend on the dependence structure.

**Follow-up:** What extra information beyond the marginal PMFs is needed to determine  $\text{Var}(X + Y)$ ?

# Wrap-up

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## Conditional expectation

- $\mathbb{E}[X \mid Y = y]$  is the average of  $X$  under the conditional PMF
- The total expectation theorem states

$$\mathbb{E}[X] = \sum_{y: p_Y(y) > 0} \mathbb{E}[X \mid Y = y] p_Y(y).$$

## Independence and expectations

- Independence means  $p_{X,Y}(x, y) = p_X(x)p_Y(y)$  for every  $(x, y)$
- Under independence,  $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$

## Sums of independent random variables

- Means add without independence; variances add for independent variables
- For  $S \sim \text{Binomial}(n, p)$ :  $\mathbb{E}[S] = np$ ,  $\text{Var}(S) = np(1 - p)$

**Suggested reading:** [BT08, Ch. 2.6–2.7]

**Next lecture:** Continuous distributions: PDFs, CDFs, uniform and exponential models

# References

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Dimitri Bertsekas and John N Tsitsiklis.

*Introduction to probability*, volume 1.

Athena Scientific, 2nd edition, 2008.